

MOHAMED THANVEER G

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PROFESSIONAL SUMMARY

High-Tech Manufacturing Engineer with **3 years** of industrial experience delivering **Industry 4.0**, **computer vision**, and **digital manufacturing** solutions in high-volume, precision production environments. MSc candidate in Mechanical Engineering (**TU Delft**, High-Tech Engineering — Micro & Nano) building on a mechanical engineering foundation and postgraduate training in **data analytics**. Proven record applying **Python-based machine vision**, **digital-twin** modelling, and **process automation** to cut defect rates, raise throughput, and improve plant-wide visibility for **operational excellence**. Skilled in translating manufacturing challenges into scalable, data-driven engineering solutions across cross-functional teams. Seeking to apply process optimisation, automation, and computer vision expertise to **precision engineering** and **semiconductor equipment manufacturing**.

EDUCATION

Technische Universiteit Delft (TU Delft) — MSc Mechanical Engineering, High-Tech Engineering | Delft, Netherlands | Aug 2026 – Present
Specialisation: Micro & Nano Engineering | Focus: Precision Engineering, Semiconductor Manufacturing Technologies, Mechatronics, Robotics & Automation

Great Lakes Institute of Management — Post Graduate Diploma in Management, Data Analytics | Chennai, India | Jul 2024 – Jul 2026
GPA: 4.0/4.0 | Pursued part-time alongside full-time engineering role | Focus: Business Analytics, Machine Learning, Statistical Modelling, Data Visualisation

Anna University, College of Engineering Guindy — B.E. Mechanical Engineering | Chennai, India | Jul 2019 – Jun 2023
GPA: 9.29/10 | Focus: Process Planning & Cost Estimation, Manufacturing Technology, Metrology, Topology Optimisation, Generative Design

WORK EXPERIENCE

Ashok Leyland Limited — Deputy Manager, Project Planning & Digital Manufacturing | Chennai, India | Jul 2023 – Aug 2026

- Engineered a **Python/OpenCV** machine-vision system for real-time process confirmation, cutting field defects by **96%** (2,200 → 90 PPM) and scaling automated quality inspection across **103 production lines**.
- Led an enterprise **digital-twin** programme across manufacturing operations, digitally replicating **24,000+ sqm** of facility using **LiDAR** scanning and 3D modelling to optimise layout, material flow, and capacity planning.
- Designed a **Python/OpenCV** vision-based **poka-yoke** system for a high-mix assembly line, eliminating variant-mismatch defects across two product platforms and improving first-time-right quality.
- Implemented an **Industry 4.0** energy-monitoring platform integrating smart meters, **PLCs**, and analytics dashboards, achieving **100% plant-wide power visibility** and driving KPI-based sustainability improvements.
- Automated in-plant logistics with an **RFID-enabled** tracking system integrated with **SAP S/4HANA** analytics, reducing **transport turnaround time by 15%**.
- Strengthened quality systems through **FMEA**, **Control Plans**, and **IATF 16949** compliance protocols, and digitised shop-floor instructions via a configuration-aware **Work Instruction System** to improve first-time-right execution and audit readiness.
- Directed capability, capacity, and facility planning; managed **budgets**, **purchase orders**, and **payment advices** alongside RFQ/RFP processes and vendor negotiations; drove cross-functional continuous improvement using **Lean** and **Six Sigma** methodologies.

Atum Robotics — Mechanical Design Intern | Chennai, India | May 2022 – Aug 2022

- Designed a planetary-gear-driven mobility mechanism (**Fusion 360**) for a social robot; applied **topology optimisation** to 3D-printed components and conducted structural load simulation to improve strength-to-weight ratio and manufacturability.

KEY PROJECTS

- Chassis Line Automation** — deployed **autonomous mobile robots (AMRs)** for line-side material feeding, removing manual dependency and increasing assembly-line capacity. **Eliminated recurring monthly rental-forklift and associated manpower costs.**
- Digital PLM Transformation (3DExperience)** — unified change management, **MBOM**, and lifecycle processes onto a single **PLM** platform, improving engineering data traceability across production sites.
- Warehouse Management System (WMS) Pilot** — engineered inbound logistics workflows for **12,000+ part variants**, minimising stock mismatch and financial write-offs while improving **material throughput by 13%**

SKILLS

Programming & Software Development: Python, Java, Kotlin, Next.js, Flutter, Android Studio

AI, Computer Vision & Data Analytics: OpenCV, Machine Learning, Statistical Modelling, Power BI, Tableau, JMP, Minitab, Statistical Process Control (SPC), Design of Experiments (DOE)

Manufacturing, Automation & Industry 4.0: Digital Twin, Industrial IoT, Smart Factory, HMI, Robotics, AMR/AGV, Lean Manufacturing, Six Sigma, Work Instruction System (WIS), Continuous Improvement, Operational Excellence

Quality & Compliance: FMEA, IATF 16949, Control Plans (CP), Root Cause Analysis

CAD, Simulation & Precision Engineering: CATIA/3DExperience, Creo, SolidWorks, Fusion 360, Siemens NX, AutoCAD, Ansys, DEFORM, Topology Optimisation, Generative Design

Project, Process & Procurement Management: SAP S/4HANA, PLM/MBOM, Budget Planning, Purchase Orders (PO), RFQ/RFP, Vendor Negotiation, Capacity Planning, Agile Project Management

PUBLICATIONS & AWARDS

Publication: “Combined Influence of Depth of Cut and Feed Rate on the Performance of Tools with Different Cutting Radius in Machining Ti6Al4V” — *Procedia CIRP (Elsevier)*, peer-reviewed, CIRP-HPC 2026. DOI: [10.1016/j.procir.2026.03.019](https://doi.org/10.1016/j.procir.2026.03.019)

Award: Reward for Individual's Search for Excellence (RISE) — *Ashok Leyland*, May 2025